

Charger (BQ24250) — Working Document

Digital Audio Player · Rev 6

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1. Part Identification

Field	Value
Manufacturer / MPN	Texas Instruments — BQ24250RGET
AL-PN	#1-054
Package	24-VQFN (4 × 4)
Function	3 MHz switch-mode Li-ion charger with PowerPath

2. Design Requirements

Requirement	Value
Input	USB-C, 5 V
Battery	1S Li-Po, 3.0 – 4.2 V, 1850 mAh
Charge current	500 mA (0.27C)
Input current limit	1 A
VIN_DPM	4.488 V
Switching frequency	3 MHz (2.7 – 3.3 MHz)
Charge temperature window	TS network, 0 – 60 °C
System peak (S4)	385 mA at 3.7 V (Power Budget Rev 4)

Constraints: Low noise · Small footprint

3. Pin Configuration

Pin	Name	Connection
1	CE	Default low (charging enabled); firmware can disable
2	EN1	10 kΩ pull-up to +3V3_DIG
3	EN2	10 kΩ pull-down to GND
4	AGND	Ground
5	SDA	10 kΩ pull-up to +3V3_DIG
6	SCL	10 kΩ pull-up to +3V3_DIG
7	STAT	Charge LED (green) + 1 kΩ from SYS
8	INT	10 kΩ pull-up to +3V3_DIG
9	TS	R2 from LDO, R3 + NTC to AGND
10	ISET	RISET to AGND
11, 12	BAT	BAT+, CBAT
13, 14	SYS	SYS, CSYS bank
15, 16	PGND	GND

Pin	Name	Connection
17, 18	SW	L, CBOOT
19	IN	VUSB (filtered), CIN, VDPM divider
20	PMID	CPMID to GND
21	BOOT	CBOOT to SW
22	ILIM	RILIM to AGND
23	VDPM	Divider tap
24	LDO	CLDO, top of TS network
25	Thermal pad	GND

4. External Components

Ref	Value	Rating / package	MPN	AL-PN
L	1.0 μ H $\pm$ 20%	2 A I_{sat} , 96 m Ω , 0505	TDK TFM141206BLE-1R0MTCA	#1-063
CBAT, CLDO, CPMID	1 μ F	16 V X6S 0402	Murata GRM155C81C105KE11D	#1-058
CBOOT	33 nF	16 V X7R 0402	Murata GRM155R71C333KA01D	#1-059
CIN (C3)	2.2 μ F	25 V X7R 0603	Murata GRM188Z71E225KE43D	#1-070
CSYS1, CSYS2	22 μ F	25 V X7S 1210	Murata GCM32EC71E226KE36L	#1-073
CSYS3	0.1 μ F	50 V X7R 0402	Murata GRM155R71H104KE14D	#1-048
RISET	500 Ω 0.1%	0603	YAGEO RT0603BRC07500RL	#1-064
RILIM	270 Ω 1%	0603	YAGEO RC0603FR-07270RL	#1-065
R_DPM1 / R_DPM2	274 k Ω / 100 k Ω 1%	0603	Generic	—
R2 (TS)	5 k Ω 1%	0402	YAGEO RC0402FR-075KL	#1-060
R3 (TS)	9.88 k Ω 0.5%	0402	YAGEO RT0402DRD079K88L	#1-061
NTC	10 k Ω , β 4000 K	On cell body	—	—
F1	120 Ω @ 100 MHz	3 A, 0805	Würth 742792023	#1-068
C1, C2 (filter)	4.7 μ F	16 V X7R 0805	Murata GRM21BR71C475KE51L	#1-071
R1 (filter damping)	0.5 Ω 1%	1/4 W 0805, pulse-rated metal film	YAGEO PE0805FRF7W0R5L	#1-097
D1	SMAJ7.0A TVS	SMA	Diodes Inc. SMAJ7.0A-13-F	#1-049
D2, D3	2-ch ESD, 5.5 V WM, 1.5 pF — D2: D+ / D—, D3: CC1 / CC2	SC-70-3	TI TPD2E2U06QDCKRQ1	#1-099
J1	USB-C receptacle	USB 2.0	GCT USB4085-GF-A	#1-069
J	Battery connector, 3-pos	5.08 mm TH	Phoenix Contact 1786417	#1-062
R_CC1, R_CC2	5.1 k Ω	0402	Generic	—
Pull-ups / pull-down	10 k Ω ($\times$ 5 / $\times$ 1)	0603	Generic	—

5. Equations and Results

Inductor ripple (V_{IN} 5 V, V_{BAT} 3.0 V, f_s 2.7 MHz, L 1.0 μ H)

$$\Delta I_L = \frac{V_{BAT}(V_{IN} - V_{BAT})}{V_{IN} f_s L} = 0.444 \text{ A}$$

Inductor peak current (Load = $I_{SYS} + I_{CHG}$)

$$I_{PEAK} = I_{Load} + \frac{\Delta I_L}{2}$$

Case	I_avg	I_PEAK
Charging, idle	0.500 A	0.722 A
Charging + S2 playback	0.637 A	0.859 A
Charging + S4 peak	0.885 A	1.107 A (vs 2 A I_sat)

Charge current

$$R_{ISET} = \frac{250}{I_{FC}} = \frac{250}{0.5} = 500 \Omega$$

Input current limit

$$R_{ILIM} = \frac{270}{I_{IC}} = \frac{270}{1.0} = 270 \Omega$$

Input DPM threshold

$$V_{IN_DPM} = 1.2 \frac{R_1 + R_2}{R_2} = 1.2 \times \frac{374}{100} = 4.488 \text{ V}$$

TS thresholds (ratiometric to V_{LDO} ; 10 k Ω β 4000 K NTC)

Threshold	V_TS / V_LDO	Temperature
Cold	60%	0 °C
Cool	56.5%	$\approx$ 13 °C
Warm	38.3%	$\approx$ 47 °C
Hot	30%	60 °C

Input current limit straps (BQ24250 Table 1)

EN2	EN1	Input limit	V_IN_DPM
0	1	External R_ILIM (1 A)	External divider (4.488 V) — operating state
0	0	500 mA	4.36 V — default while +3V3_DIG is down

Filter damping resistor pulse check (hot plug, $C = 4.7 \mu\text{F}$)

$$E = \frac{1}{2}CV^2 = \frac{1}{2} \times 4.7 \mu\text{F} \times 5^2 = 59 \mu\text{J}, \quad \tau = 0.5 \Omega \times 4.7 \mu\text{F} = 2.4 \mu\text{s}, \quad P_{pk} \leq \frac{5^2}{0.5} = 50 \text{ W}$$

PE series 0805 permits $\approx 200 \text{ W}$ single pulse at $100 \mu\text{s}$ (limit rises for shorter pulses): $\geq 4\times$ margin.

6. Important Values

Item	Value
Input filter insertion loss	-83 dB at 3 MHz
Filter output impedance peak	0.46Ω with damping (46Ω without); limit 6.25Ω
SYS ripple (sim)	2.4 mV pp
Inductor DCR loss (charging + playback)	$\approx 39 \text{ mW}$
Inductor peak, worst case	1.107 A (55% of $2 \text{ A } I_{\text{sat}}$)
Battery FET R_ON(BAT-SYS)	$30 \text{ m}\Omega$ typ / $40 \text{ m}\Omega$ max
IC current limit	2.6 A min
R1 pulse margin	$\geq 4\times$ (50 W worst case vs $\approx 200 \text{ W}$ at $100 \mu\text{s}$, 0805 PE)
VBUS capacitance	$11.6 \mu\text{F}$ nominal ($C219 + C220 + C_{\text{IN}}$); $< 10 \mu\text{F}$ effective at 5 V — within USB 2.0 limit
Grounding	AGND and GND resolved to a single ground

#	Problem
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7. Open Problems

#	Problem
1	I2C may only work while USB is present (powered from IN) — verify before driver
2	NTC return shares battery return on J pin 3
3	Battery connector height vs 13.2 mm enclosure
4	Cell limits charging to 0.2C (380 mA) at 0 – 15 °C and forbids charging above 50 °C; charger runs 500 mA until ≈ 13 °C and charges to 60 °C
5	Re-simulate SYS ripple with final $2 \times 22 \mu\text{F} + 0.1 \mu\text{F}$ bank

8. Revision Log

Rev	Date	Change
0	2026-09-19	Initial issue: part selection, sizing, calculations, TS network, input filter and SYS ripple simulations.
1	2026-09-19	Reviewed against schematic BQ-CHARGER-V1; pin connections and schematic findings added.
2	2026-09-21	Low noise and small footprint constraints added; cell charge-temperature mismatch added; condensed to essentials.
3	2026-09-22	S4 system peak 462 → 385 mA (0.6 V RMS max output): inductor peak 1.184 → 1.107 A.
4	2026-09-23	Schematic review: R1 to 0805 pulse-rated PE (#1-097); USB D+/D– ESD (#1-099); EN1/EN2 strap verified. Closed: R1SET value, STAT, CE default, VDPM resistor, D+/D– routing, R1 pulse rating.
5	2026-09-23	Second TPD2E2U06 on CC1 / CC2. Closed: CC protection.
6	2026-09-23	Closed: AGND / GND, VBUS capacitance ($< 10 \mu\text{F}$ effective).